

UNITED STATES CORRECTIONAL FACILITIES MASTER DATABASE

*Comprehensive Ingestion, Normalization, Deduplication, and Validation
Methodology Report*

Publication Date: August 20, 2026 | **Coverage:** All 50 States, DC, PR, GU, VI, MP (6,788 Facilities) |
Version: 2.4 (Audited & Verified)

1. Executive Summary

This methodology report establishes the technical architecture, data provenance, cleaning rules, and deduplication logic utilized to build the United States Correctional Facilities Master Database. The resulting master dataset provides researchers, government agencies, and policy analysts with a unified, standardized repository of 6,788 physical correctional institutions operating across all 50 US states, the District of Columbia, and five US territories (Puerto Rico, Guam, US Virgin Islands, and Northern Mariana Islands).

Metric Description	Aggregated Total	Completeness / Verification
Total Unique Facilities	6,788	100.0% Unique Primary IDs
Facilities with Valid GPS Coordinates	6,788	100.0% Geocoded (WGS84)
Total Rated Bed Capacity (Design)	2,411,708 beds	Official Agency Rated Counts
Total Reported Inmate Population	2,069,547 inmates	Point-in-Time Census Data
Geographic Jurisdictions Covered	55 Jurisdictions	50 States + DC + PR, GU, VI, MP

2. Data Sources & Ingestion Architecture

To ensure comprehensive coverage across federal, state, county, and private correctional operations, the aggregation pipeline programmatically queries primary government data repositories:

- **1. DHS Homeland Infrastructure Foundation-Level Data (HIFLD):** Direct REST query against the national Critical Infrastructure Prison Points FeatureServer layer (services4.arcgis.com). Ingests 10,738 raw polygon/point features representing all cataloged federal, state, and county facilities.
- **2. DOJ Federal Bureau of Prisons (BOP) National Directory:** Ingests live structured metadata from the official BOP facility management endpoint (bop.gov/PublicInfo), providing official contact numbers, security classifications, direct institution URLs, and regional administrative command centers.

3. Cleaning, Normalization & Deduplication Methodology

Raw government correctional datasets contain significant fragmentation, duplicate multi-part GIS records, and sentinel placeholders. The pipeline enforces rigorous cleaning and enrichment algorithms:

- **Multi-Part Feature Deduplication:** HIFLD polygon boundaries frequently export multiple geometry centroids for a single campus, resulting in 4,001 redundant rows. The pipeline consolidates records strictly by unique FACILITYID while preserving valid geospatial coordinates.
- **Type-Guarded & Camp-Parity BOP Entity Matching:** To prevent false positives across both county facilities and intra-federal complexes (e.g. Beaumont, Atlanta, Coleman), the pipeline enforces strict federal type guards, separates administrative entities (RRM, Regional Offices, FCC complexes) into standalone records, and enforces camp-to-camp matching parity so satellite camps do not steal parent institution URLs (e.g. USP Beaumont correctly receives /institutions/bmp/ while FCI Beaumont Low receives /institutions/bml/).
- **Preservation of Zero-Population Data:** Valid zero-population counts for brand-new, temporarily unpopulated, or specialized intake facilities (625 records) are retained as 0, while negative sentinels (-999, -1, 99999) are scrubbed to null.
- **Smart Typography & Acronym Normalization:** Addresses, cities, and facility names are standardized to Title Case while strictly preserving uppercase acronyms (USP, FCI, ADX, MDC, FDC, FMC, BOP, DOC, SCI, ASPC, CCFW) and Scottish/Irish prefixes (McDuffie, McCreary, McKean, O'Brien).
- **FIPS & Geographic Imputations:** Corrects upstream FIPS typos (e.g. Pickens County AL '10107' -> '01107') and provides complete county FIPS mapping for standalone federal facilities and territories (Guam FIPS 66010, US Virgin Islands FIPS 78010, Saipan FIPS 69110).

4. Master Directory Distribution by Jurisdiction

The master dataset categorizes facilities into six standardized governmental authority tiers:

Jurisdiction Level	Facility Count	Percentage of Dataset
County / Local Jails	3,960	58.3%
State DOC Facilities	2,273	33.5%

Federal (BOP & USMS)	308	4.5%
Municipal / City Lockups	184	2.7%
Multi-Jurisdiction Facilities	36	0.5%
Not Specified (Tribal / Contract / Unrecorded)	27	0.4%

5. Known Data Limitations & Upstream Source Anomalies

Researchers, auditors, and policy analysts should take note of the following known characteristics and upstream data anomalies present in primary government source records:

- **Point-in-Time Census Variations:** Reported inmate populations reflect point-in-time census snapshots provided during individual state and federal reporting cycles rather than live real-time daily headcounts.
- **Upstream Population vs. Capacity Outliers:** There are 8 specific county/state facilities where reported population exceeds 3x rated design capacity (e.g., Woodman State Jail TX reporting population 6,478 vs design capacity 900; Fulton County Jail IN reporting 133 vs 35; Burke County Jail NC reporting 262 vs 66). These represent known upstream DHS HIFLD data artifacts preserved as recorded by the source agency.
- **Campus Co-Location of Distinct Agencies:** A small number of municipal and state facilities share identical street addresses or GPS coordinates due to co-located campuses (e.g., county sheriff dispatch and municipal jail in the same civic center, or psychiatric hospital and detention unit on one state hospital campus). These are preserved as separate records with unique primary IDs.

6. Standardized Data Dictionary (20 Master Fields)

Both the master CSV and Excel spreadsheets feature 20 fully standardized columns. The table below defines each programmatic field name and its corresponding Excel column header:

Display Column Header	CSV Field (snake_case)	Description & Type
Facility ID	facility_id	String: Unique primary identifier (HIFLD ID or BOP Code).
Facility Name	facility_name	String: Standardized institution title case name.
Jurisdiction	jurisdiction	String: Level of authority (Federal, State, County, etc.).

Facility Classification	facility_type	String: Prison, Jail, Juvenile, Medical/Psych, Reentry.
Security Level	security_level	String: Maximum, Close, Medium, Minimum, Administrative.
Operational Status	operational_status	String: Open, Closed, or Not Available.
Street Address	street_address	String: Physical street address of facility.
City	city	String: City or municipality.
State	state	String: 2-letter postal code (50 states + 5 territories).
ZIP Code	zip_code	String: 5-digit ZIP code with preserved leading zeros.
County	county	String: County, parish, or borough name.
County FIPS	county_fips	String: 5-digit FIPS code with preserved leading zeros.
Phone Number	phone_number	String: Standardized (XXX) XXX-XXXX phone number.
Website	website	String: Official governing portal or institution URL.
Latitude	latitude	Float: WGS84 Decimal Degrees Latitude (North).
Longitude	longitude	Float: WGS84 Decimal Degrees Longitude (West/East).
Design Capacity	design_capacity	Integer: Official rated design bed count.
Population	population	Integer: Reported inmate population count.
Gender	gender	String: Male, Female, Co-ed, or Not

		Specified.
Data Source	data_source	String: Primary origin source agency.